DEFENCE JOURNEY

CDS Elementary Mathematics Notes

Arithmetic, algebra, geometry, mensuration and trigonometry for the CDS Maths paper

CDS | MATHS

The Elementary Mathematics paper has **100 questions for 100 marks** (1 mark each, 1/3 negative marking) and is needed for IMA, INA and AFA. OTA candidates do not write it. The level is Class 8 to 10, so speed and accuracy matter more than difficulty. Learn the formulas and shortcuts here and practise with a timer.

Section	Topics	Weight
Arithmetic	Number system, HCF/LCM, percentage, profit and loss, interest, ratio, time and work, speed, averages, mixtures	~40%
Algebra	Identities, linear and quadratic equations, indices, surds, polynomials	~15%
Geometry	Lines, triangles, circles, quadrilaterals, polygons, coordinate basics	~15%
Mensuration	Area, perimeter, surface area and volume	~15%
Trigonometry	Ratios, identities, heights and distances	~10%
Statistics	Mean, median, mode, charts	~5%

1. Number System

- Natural numbers: 1, 2, 3 … Whole numbers: 0, 1, 2 … Integers: … −2, −1, 0, 1, 2 … Rational numbers: p/q (q ≠ 0). Irrational numbers: √2, π.
- **Prime numbers** below 50: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47. 2 is the only even prime. 1 is neither prime nor composite.
- **Divisibility:** by 2 (last digit even), 3 (digit sum divisible by 3), 4 (last two digits), 5 (ends in 0 or 5), 6 (by 2 and 3), 8 (last three digits), 9 (digit sum divisible by 9), 11 (difference of sums of alternate digits is 0 or a multiple of 11).
- **HCF × LCM = product of the two numbers.** HCF is the largest number dividing all. LCM is the smallest number divisible by all. HCF of fractions = HCF of numerators / LCM of denominators.
- The largest number that divides a, b, c leaving remainders r_1, r_2, r_3 is the HCF of $(a - r_1)$, $(b - r_2)$, $(c - r_3)$.
- Unit digit of powers repeats in a cycle. For 2: 2, 4, 8, 6. For 3: 3, 9, 7, 1. For 7: 7, 9, 3, 1. For 4, 9: two-step cycle. 0, 1, 5, 6 never change.
- Squares to remember: $11^2 = 121$, $12^2 = 144$, $13^2 = 169$, $14^2 = 196$, $15^2 = 225$, $16^2 = 256$, $17^2 = 289$, $18^2 = 324$, $19^2 = 361$, $25^2 = 625$. Cubes: $5^3 = 125$, $6^3 = 216$, $7^3 = 343$, $8^3 = 512$, $9^3 = 729$.
- Number of factors of $N = p^a q^b r^c$ is $(a+1)(b+1)(c+1)$.

2. Percentage, Profit and Loss, Discount

Fraction	Percent	Fraction	Percent	Fraction	Percent
1/2	50%	1/6	16.67%	1/9	11.11%
1/3	33.33%	1/7	14.28%	1/10	10%
1/4	25%	1/8	12.5%	1/12	8.33%
1/5	20%	2/3	66.67%	3/8	37.5%

- Percentage = (part/whole) × 100. Increase or decrease % = (change/original) × 100.
- **Successive changes** of a% and b%: net change = a + b + (ab/100). (Use a negative sign for a decrease.)
- Profit = SP − CP. Profit % = (Profit/CP) × 100. Loss % = (Loss/CP) × 100. **SP = CP × (100 + P%)/100.**
- Discount = Marked price − Selling price. Discount % = (Discount/MP) × 100. Two successive discounts a% and b% = a + b − (ab/100).
- If two articles are sold at the same price, one at a gain of x% and the other at a loss of x%, there is **always a loss** of $(x/10)^2$ %.

3. Simple and Compound Interest

- Simple interest: **SI = PRT/100**. Amount A = P + SI.
- Compound interest (yearly): **$A = P(1 + R/100)^n$**, CI = A − P. For half-yearly, use rate R/2 and time 2n.
- **CI − SI for 2 years = $P(R/100)^2$**.
- A sum becomes double in T years at simple interest, then at rate R, T = 100/R years. It becomes n times in 100(n − 1)/R years.

4. Ratio, Proportion, Mixtures and Partnership

- If a : b = c : d, then **ad = bc**. Compounded ratio of a : b and c : d is ac : bd.
- **Alligation:** (Cheaper quantity) : (Dearer quantity) = (d − m) : (m − c), where c is the price of the cheaper, d is the price of the dearer and m is the mean price.
- Direct variation: x ∝ y means x/y is constant. Inverse variation: x ∝ 1/y means xy is constant.
- Partnership: profit is shared in the ratio of (investment × time).
- Ages: if the ratio now is a : b and the difference is d, then the ages are in the same ratio multiplied by d/(b − a).

5. Time and Work, Pipes

- If A does a job in **a** days and B in **b** days, together they take **ab/(a + b)** days.
- Work = rate × time. If A is x times as efficient as B, A takes 1/x of B's time.
- **Pipes and cisterns:** an inlet pipe fills (positive work), an outlet empties (negative work). Net rate = sum of rates.
- $M_1D_1/W_1 = M_2D_2/W_2$ (men × days / work is constant).

6. Time, Speed and Distance

- **Speed = distance/time.** To convert km/h to m/s, multiply by 5/18. To convert m/s to km/h, multiply by 18/5.
- Average speed over a journey = total distance/total time. For equal distances at speeds x and y: **average speed = 2xy/(x + y)**.
- Relative speed: opposite directions add the speeds; same direction subtract the speeds.
- **Trains:** time to cross a pole or a man = length of train/speed. Time to cross a platform = (length of train + length of platform)/speed. For two trains: add or subtract speeds and add the lengths.
- **Boats and streams:** downstream speed = u + v; upstream speed = u − v (u is the boat's speed in still water, v is the stream speed). u = (downstream + upstream)/2, v = (downstream − upstream)/2.
- If the speed is in the ratio a : b, the time taken for the same distance is in the ratio b : a.

7. Averages

- Average = sum of observations/number of observations.
- Average of the first n natural numbers = (n + 1)/2. Average of the first n even numbers = n + 1. Average of the first n odd numbers = n.
- If a new member joins and the average changes by d, the new member's value = old average + (n + 1)d, where n is the old number of members.
- Average of two groups = $(n_1a_1 + n_2a_2)/(n_1 + n_2)$.

8. Algebra

- **Identities:** $(a + b)^2 = a^2 + 2ab + b^2$. $(a - b)^2 = a^2 - 2ab + b^2$. $a^2 - b^2 = (a + b)(a - b)$.
- $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$. $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$. $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$.
- If a + b + c = 0, then $a^3 + b^3 + c^3 = 3abc$.
- If $x + 1/x = k$, then $x^2 + 1/x^2 = k^2 - 2$.
- **Indices:** $a^m \times a^n = a^{m+n}$. $a^m/a^n = a^{m-n}$. $(a^m)^n = a^{mn}$. $a^0 = 1$. $a^{-n} = 1/a^n$.
- **Surds:** $\sqrt{a} \times \sqrt{b} = \sqrt{(ab)}$. $(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b$. To rationalise $1/(\sqrt{a} + \sqrt{b})$, multiply by $(\sqrt{a} - \sqrt{b})/(\sqrt{a} - \sqrt{b})$.
- **Quadratic equation** $ax^2 + bx + c = 0$: roots = $[-b \pm \sqrt{(b^2 - 4ac)}]/2a$. Sum of roots = $-b/a$. Product of roots = c/a.
- **Linear equations:** $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$ have a unique solution if $a_1/a_2 \neq b_1/b_2$; no solution if the ratios of coefficients are equal but not equal to c_1/c_2.
- **Remainder theorem:** if f(x) is divided by (x − a), the remainder is f(a). If f(a) = 0, (x − a) is a factor.

9. Geometry

- Sum of angles of a triangle = 180°. An exterior angle = sum of the two opposite interior angles.
- **Pythagoras:** in a right-angled triangle, $\text{hypotenuse}^2 = \text{base}^2 + \text{perpendicular}^2$. Triples: (3, 4, 5), (5, 12, 13), (8, 15, 17), (7, 24, 25).

- **Similar triangles:** corresponding angles are equal and sides are in proportion. The ratio of areas = square of the ratio of corresponding sides.
- **Basic Proportionality (Thales) theorem:** a line parallel to one side of a triangle divides the other two sides in the same ratio.
- The **centroid** (meeting of medians) divides each median in the ratio 2 : 1. The **incentre** is the centre of the inscribed circle (angle bisectors). The **circumcentre** is the centre of the circumscribed circle (perpendicular bisectors). The **orthocentre** is the meeting of altitudes.
- **Circle:** the angle at the centre is twice the angle at the circumference on the same arc. The angle in a semicircle is 90°. Tangents from an external point are equal in length. A tangent is perpendicular to the radius at the point of contact.
- **Cyclic quadrilateral:** the opposite angles add up to 180°.
- **Polygons:** sum of interior angles = $(n - 2) \times 180°$. Each interior angle of a regular polygon = $(n - 2) \times 180°/n$. Sum of exterior angles = 360°. Number of diagonals = $n(n - 3)/2$.
- Parallel lines cut by a transversal: corresponding angles are equal, alternate angles are equal, co-interior angles are supplementary.

10. Mensuration

Shape	Area / Surface area	Perimeter / Volume
Rectangle	$l \times b$	$2(l + b)$
Square	a^2 $(= d^2/2)$	$4a$
Triangle	$\frac{1}{2} \times$ base $\times$ height	$a + b + c$
Equilateral triangle	$(\sqrt{3}/4)a^2$	$3a$
Parallelogram	base $\times$ height	$2(a + b)$
Rhombus	$\frac{1}{2} \times d_1 \times d_2$	$4a$
Trapezium	$\frac{1}{2}(a + b) \times h$	sum of all sides
Circle	πr^2	$2\pi r$ (circumference)
Cuboid	$TSA = 2(lb + bh + hl)$	$V = lbh$. Diagonal $= \sqrt{(l^2 + b^2 + h^2)}$
Cube	$TSA = 6a^2$, $LSA = 4a^2$	$V = a^3$. Diagonal $= a\sqrt{3}$
Cylinder	$CSA = 2\pi rh$, $TSA = 2\pi r(r + h)$	$V = \pi r^2 h$
Cone	$CSA = \pi rl$, $TSA = \pi r(r + l)$, $l = \sqrt{(r^2 + h^2)}$	$V = \frac{1}{3}\pi r^2 h$
Sphere	$4\pi r^2$	$V = (4/3)\pi r^3$
Hemisphere	$CSA = 2\pi r^2$, $TSA = 3\pi r^2$	$V = (2/3)\pi r^3$

- Use $\pi = 22/7$ or 3.14 as the question suggests. Convert all measurements to the same unit before using a formula.
- If the radius of a circle is made n times, the area becomes n^2 times. If the side of a square is increased by x%, the area increases by $[2x + x^2/100]\%$.
- Melting and recasting: the volume stays constant.

11. Trigonometry

θ	0°	30°	45°	60°	90°
sin θ	0	1/2	1/√2	√3/2	1
cos θ	1	√3/2	1/√2	1/2	0
tan θ	0	1/√3	1	√3	not defined

- sin θ = perpendicular/hypotenuse, cos θ = base/hypotenuse, tan θ = perpendicular/base. cosec, sec, cot are their reciprocals.
- $\sin^2\theta + \cos^2\theta = 1$. $1 + \tan^2\theta = \sec^2\theta$. $1 + \cot^2\theta = \text{cosec}^2\theta$.
- $\sin(90° - \theta) = \cos\theta$, $\cos(90° - \theta) = \sin\theta$, $\tan(90° - \theta) = \cot\theta$.
- **Heights and distances:** draw a figure. Angle of elevation is measured upward and angle of depression downward. If the angle of elevation is 45°, height = distance. For 30° with height h, distance = h√3. For 60°, distance = h/√3.

12. Statistics

- **Mean** = sum/number. **Median:** middle value when arranged in order (for an even number of values, the average of the two middle ones). **Mode:** most frequent value.
- Mode = 3 Median − 2 Mean.
- **Range** = highest − lowest. Standard deviation measures spread.
- If every observation is increased by k, the mean increases by k and the standard deviation does not change. If every observation is multiplied by k, both are multiplied by k.
- In a pie chart, the angle for an item = (item/total) × 360°.

Exam Strategy

- Total time is 2 hours for 100 questions. Spend no more than 1 minute per question. Skip any question that takes longer and come back.
- Use **option substitution** and **approximation** where possible. The answer choices are usually far apart.
- Memorise tables up to 20, squares up to 30 and cubes up to 15.
- Practise 40 questions daily on one topic, then one full mock test every week.
- Keep a notebook of formulas and your repeated mistakes.